

Project Report on Spam Email/SMS Classifier using Machine Learning

An Project Report submitted
in partial fulfillment of the requirements for the degree
of

B. Tech. (CSE)

Submitted by

Sumit Oinam (242025033)

B.Tech. (CSE), 5th Semester

Under the guidance of

Roshan Dey

Program Coordinator

Project Title

Spam Email/SMS Classifier using Machine Learning

1. Project Description

The Spam Email/SMS Classifier is a Machine Learning-based application that predicts whether an email or SMS message is **Spam** or **Not Spam (Ham)**. The system is trained using a labeled SMS dataset and uses Natural Language Processing (NLP) techniques to convert text into numerical features before classification.

The project is developed using **Python**, **Scikit-learn**, and **Streamlit**. A trained Machine Learning model is deployed as a web application so that users can enter any message and instantly receive a prediction.

2. Project Objectives

- Detect spam messages automatically.
 - Reduce unwanted spam messages.
 - Demonstrate the practical application of Machine Learning.
 - Provide an easy-to-use web interface for prediction.
-

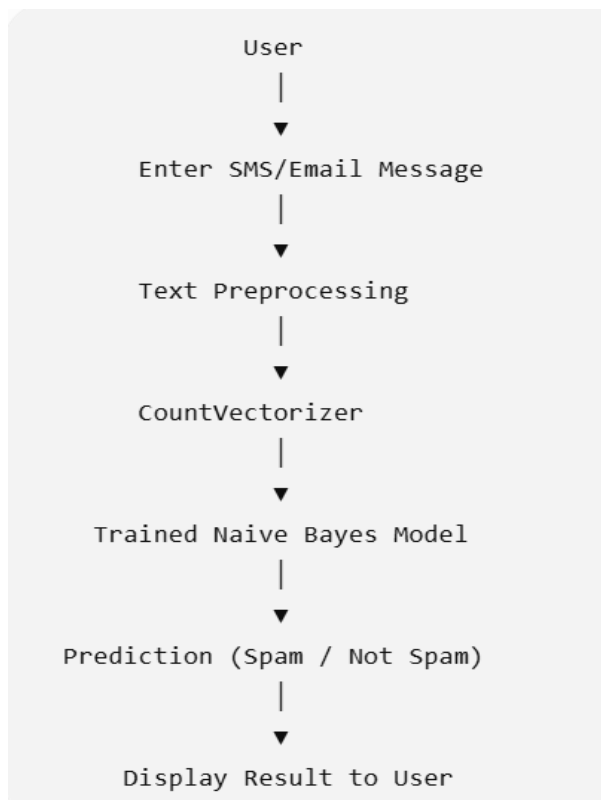
3. Key Features (Functionalities)

- Classifies SMS and email messages.
 - User-friendly web interface.
 - Fast prediction using a trained Machine Learning model.
 - Supports both spam and non-spam detection.
 - Deployed online using Streamlit Cloud.
 - Uses Natural Language Processing (NLP).
-

4. Tech Stack

Component	Technology Used
Programming Language	Python 3.13
IDE	Visual Studio Code
Machine Learning	Scikit-learn
Data Processing	Pandas
Feature Extraction	CountVectorizer
Classification Algorithm	Multinomial Naive Bayes
Web Framework	Streamlit
Version Control	Git & GitHub
Deployment	Streamlit Community Cloud

5. Project Architecture



6. Project Structure

spam_email_sms_classifier/

|

|— .gitignore

|— README.md

|— app.py

|— model.pkl

|— requirements.txt

|— spam.csv

|— train_model.py

|— vectorizer.pkl

7. Project Pipeline

Step Description

- 1 Load the SMS Spam Collection Dataset
 - 2 Convert labels (Ham = 0, Spam = 1)
 - 3 Split the dataset into training and testing sets
 - 4 Convert text into numerical vectors using CountVectorizer
 - 5 Train the Multinomial Naive Bayes model
 - 6 Save the trained model and vectorizer
 - 7 Create a Streamlit web application
 - 8 Deploy the application using Streamlit Cloud
-

8. Installation

Clone the Repository

```
git clone https://github.com/sumitoinam14-alt/spam_email_sms_classifier.git
```

Move into the Project Folder

```
cd spam_email_sms_classifier
```

Install Required Packages

```
pip install -r requirements.txt
```

Run the Application

```
streamlit run app.py
```

9. Usage

1. Open the deployed web application.
 2. Enter an email or SMS message.
 3. Click the **Predict** button.
 4. The application displays whether the message is **Spam** or **Not Spam**.
-

10. Input and Output

Input	Output
"Congratulations! You have won ₹10,000. Click here."	Spam
"Hey, are we meeting at 6 PM?"	Not Spam
"Claim your free prize now!"	Spam
"See you tomorrow in class."	Not Spam

11. Dataset Source

Item	Details
Dataset Name	SMS Spam Collection Dataset
Source	UCI Machine Learning Repository
Total Messages	5,572
Ham Messages	4,825
Spam Messages	747

12. Machine Learning Model

Property	Value
Algorithm	Multinomial Naive Bayes
Feature Extraction	CountVectorizer
Train-Test Split	80% Training, 20% Testing

13. Results

The trained model achieved the following performance:

Metric	Value
Accuracy	96.23%
Precision (Ham)	0.96
Recall (Ham)	1.00
F1-Score (Ham)	0.98
Precision (Spam)	1.00
Recall (Spam)	0.72

Metric	Value
--------	-------

F1-Score (Spam)	0.84
-----------------	------

Classification Summary

Class	Precision	Recall	F1-Score	Support
Ham (0)	0.96	1.00	0.98	965
Spam (1)	1.00	0.72	0.84	150

14. Software Requirements

Software	Version
Windows	10/11
Python	3.13
Visual Studio Code	Latest
Git	Latest
GitHub Account	Required
Streamlit	Latest

15. Hardware Requirements

Hardware	Minimum Requirement
Processor	Intel Core i3 or above
RAM	4 GB
Storage	2 GB Free Space
Internet	Required for deployment

16. Challenges Faced

During the development of this project, several challenges were encountered:

- Understanding Machine Learning concepts for the first time.
- Installing and configuring Python libraries correctly.
- Converting text data into numerical vectors.
- Saving and loading the trained model using Pickle.
- Deploying the project successfully on Streamlit Cloud.
- Fixing file path issues during deployment.
- Organizing project files for GitHub.

These challenges were resolved through testing, debugging, and proper project organization.

17. Future Improvements

- Support multiple languages.
 - Add Deep Learning models for higher accuracy.
 - Improve spam detection using advanced NLP techniques.
 - Support attachment scanning.
 - Maintain a continuously updated spam database.
 - Develop Android and iOS applications.
 - Integrate with email services like Gmail and Outlook.
-

18. Conclusion

The **Spam Email/SMS Classifier** successfully demonstrates the use of Machine Learning for text classification. The application accurately predicts whether an email or SMS is spam using the **Multinomial Naive Bayes algorithm** and **CountVectorizer**. It achieved an overall accuracy of **96.23%**, making it reliable for real-world spam detection. The project also demonstrates the complete Machine Learning workflow, including data preprocessing, model training, evaluation, deployment, and user interaction through a Streamlit web application.

19. References

1. UCI Machine Learning Repository – SMS Spam Collection Dataset
2. Scikit-learn Documentation
3. Streamlit Documentation
4. Python Official Documentation
5. Pandas Documentation
6. GitHub Documentation